an appliance, where we control the heat of the appliance by simply putting the switch 'ON' and 'OFF' only according to the types of cloths used.

Fig. 3.1 **Non-automatic iron box**

The main parts of the iron box are listed below

a) Handle

Handle is made up of bakelite a hard type of plastic in the iron box. Bakelite is having resistive power of heat and non-conductor of electricity. A rubber tube is provided at the top of the handle, to avoid the damage of power chord while in use.

b) Top cover

Top cover is at the top of the appliance and it covers all the inner parts of the appliance. It is made of metal which is chromium plated to look attractive.

c) Stand

Stand is made up of strong iron to make the appliance to keep stand when not in use.

d) Pressure plate

Pressure plate is made up of hard cast iron and it gives weight to the appliance. Due to this, the wrinkles in the clothes are removed.

e) Mica sheet

Mica is an insulator of electricity. It is kept in top and bottom of the heating element for safety purpose. Mica is also good to resist the heat.

f) Heating element

The heating element of the iron box is made from mixed alloy of Nichrome. Usually in all the heating appliances, it is used as heating element. The heating element here used is of ribbon type.

Heating element is of two types.

1. Coiled type and
2. Ribbon type.

g) Sole plate

Sole plate is plated with chromium and made up of cast iron. The bottom of the sole plate is surface grained for smoothness.

Working principle

The power chord of the iron box is connected to the main supply. The electric energy is converted into heat due to this heating element. The heating element is having the property of high resistivity. According to the law of conservation of energy the heat energy produced is proportional to the square of current without any loss. This heat energy makes the sole plate to hot and with that the wrinkles in the cloth are removed.

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